Inverting the Lithospheric Stress Field of Central Iran: A Physics-Informed Deep Learning Approach using Sparse Kinematic Data

Research Proposal

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Abstract

Accurate estimation of the lithospheric stress field is critical for earthquake forecasting but remains challenging due to sparse *in-situ* measurements. Traditional Finite Element Models (FEMs) require complex meshing and explicit boundary conditions that are often unknown. We propose a novel **Physics-Informed Neural Network (PINN)** framework to invert for the 3D continuous stress tensor of Central Iran. Unlike previous studies that assume purely elastic rheology, our model incorporates **Maxwell Viscoelasticity** to capture time-dependent stress relaxation. Furthermore, we introduce a method to constrain the network using only **sparse GPS azimuths** (orientations), solving a highly non-linear inverse problem. This approach promises a mesh-free, physics-consistent stress map that outperforms statistical baselines (XGBoost) and standard Coulomb Stress Transfer (CST) in forecasting seismicity.

1 Scientific Rationale & Gap Analysis

Current state-of-the-art methods generally fall into two categories:

1. **Data-Driven (e.g., XGBoost)**: Excellent at finding correlations but lacks physical constraints, leading to poor generalization in data-sparse regions.
2. **Classical Physics (e.g., FEM/CST)**: Physically rigorous but limited by mesh quality and the need for unknown boundary conditions.

The Gap

Recent works (e.g., *Zhang et al., Sci. Rep. 2022*) have demonstrated PINNs for stress inversion but rely on simplified **Linear Elastic** assumptions and dense synthetic data. No study has yet applied **Viscoelastic PINNs** to invert for the stress field of the complex **Central Iranian collision zone** using real, sparse **GPS strain azimuths**.

2 Methodology

2.1 Physics-Informed Architecture

We define a deep neural network $\mathcal{N}_\theta(x, y, z, t)$ that approximates the stress tensor σ_{ij} and velocity field v_i. The Loss Function $\mathcal{L}$ minimizes the residuals of the momentum balance and constitutive equations:

$$\mathcal{L}_{total} = \mathcal{L}_{metric} + \lambda_1 \mathcal{L}_{PDE} + \lambda_2 \mathcal{L}_{BC} \tag{1}$$

where:

- $\mathcal{L}_{PDE}$ (Momentum): $\nabla \cdot \sigma + \rho g = 0$
- $\mathcal{L}_{PDE}$ (Constitutive): $\dot{\epsilon}_{ij} = \frac{1}{2\mu}\dot{\sigma}_{ij} + \frac{1}{2\eta}\sigma_{ij}$ (Maxwell Viscoelasticity)

2.2 Data Integration (The "Inverse" Problem)

We utilize the provided `kinematic_data` (GPS strain rates). Crucially, these data provide **orientations** (azimuths) rather than absolute stress magnitudes. We enforce this constraint by penalizing the misalignment between the predicted principal stress direction and the observed GPS strain azimuth:

$$\mathcal{L}_{data} = \sum_k (1 - \cos(\theta_{pred}^{(k)} - \theta_{obs}^{(k)}))^2 \tag{2}$$

2.3 Execution Roadmap

1. **Phase I (Validation)**: Implement a 2D Plane Stress model to validate the formulation against a synthetic "toy" fault interaction.
2. **Phase II (Main)**: Scale to full 3D Lithospheric model using the Central Iran velocity structure (`Pwave.3D.txt`) as a proxy for heterogeneous material properties ($\mu(x)$).

3 Validation & Benchmarking

To satisfy the rigor required by high-impact journals, we will benchmark against:

1. **Coulomb Stress Transfer (CST)**: The standard physics-based predictor.
2. **XGBoost (Spatial Susceptibility)**: The standard ML baseline.

We will demonstrate that the PINN model achieves superior log-likelihood in **hindcasting** historical seismicity (2006-2023) compared to these baselines.

4 Expected Impact

This study will provide the first-ever continuous, physics-consistent 3D stress map of Central Iran. By demonstrating the efficacy of Viscoelastic PINNs in data-sparse regions, we establish a new framework for earthquake potential imaging applicable to collision zones worldwide.